Simone **D'Ambrogio**

DPhil Candidate

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This CV is interactive Chat with it and ask anything about my research at **https://simonedambrogio.github.io/chat**

Education

University of Oxford — *Oxford, UK*
DPhil in Experimental Psychology — *2021 – 2026*
Dissertation: *Learning Efficiently in Uncertain and Structured Worlds*
Advisors: Prof. Matthew F. S. Rushworth, Prof. Laurence T. Hunt
Examiners: Prof. Thomas Akam, Prof. Nathaniel D. Daw

University of Pennsylvania — *Philadelphia, PA*
Visiting Graduate Scholar, Wharton Neuroscience Initiative — *2020*

University of Padua — *Padua, Italy*
M.Sc. in Applied Cognitive Psychology (Cum Laude) — *2018 – 2020*

Leiden University — *Leiden, Netherlands*
Erasmus+ Programme — *2018*

University of Padua — *Padua, Italy*
B.Sc. in Psychology (First Class Honours) — *2014 – 2017*

Research Interests

Decision-making · Computational modeling of cognition · Mechanistic Interpretability · Bayesian Inference · Generalization and Abstraction · Reinforcement learning

Publications

D'Ambrogio, **S.**, Fei, Z., Platt, M., & Feng, S. (2026). *From pixels to preferences: Visual salience shapes economic choices beyond gaze alone*. *PNAS (Under Review)*.

D'Ambrogio, **S.**, Grohn, J., Khalighinejad, N., Mattar, M., Hunt, L., & Rushworth, M. F. S. (2026). *Interpretable abstractions of artificial neural networks predict behavior and neural activity during human information gathering*. *Nature Neuroscience (In Press)*. https://doi.org/10.1101/2025.06.24.661282

Miyamoto*, K., **D'Ambrogio***, **S.**, Harbison, C., Eichert, N., Schüffelgen, U., Emberton, A., Sallet, J., Mars, R. B., Khalighinejad, N., & Rushworth, M. F. (2026). *Brain activity, disruption and connectivity comparisons identify origins of human metacognition in other primates* [* Equal contribution]. *Nature Human Behaviour*, 1–22. https://doi.org/10.1038/s41562-026-02473-w

Pan, D., **D'Ambrogio**, **S.**, Kingston, N., Rascu, M., Sankhe, P., Luo, S., Klein-Flügge, M. C., Mahmoodi, A., & Rushworth, M. F. S. (2025). *Construction and disruption of cognitive graphs in the human hippocampus*, 2025.08.19.664920. https://doi.org/10.1101/2025.08.19.664920

D'Ambrogio, **S.**, Werksman, N., Platt, M. L., & Johnson, E. N. (2023). *How celebrity status and gaze direction in ads drive visual attention to shape consumer decisions*. *Psychology & Marketing*, *40*(4), 723–734. https://doi.org/10.1002/mar.21772

Priolo*, G., Stablum*, F., Vacondio*, M., **D'Ambrogio**, **S.**, Caserotti, M., et al. (2023). *The robustness of mental accounting across 21 countries*. Retrieved October 18, 2025, from https://osf.io/apc26_v1

Ruggeri, K., Panin, A., ..., **D'Ambrogio**, **S.**, ..., Zupan, Z., & García-Garzon, E. (2022). *The globalizability of temporal discounting*. *Nature Human Behaviour*, *6*(10), 1386–1397. https://doi.org/10.1038/s41562-022-01392-w

Research Experience

Decision and Action Laboratory, University of Oxford *Oxford, UK*
Computational modeling, 7T fMRI, and non-invasive brain stimulation (TUS) to study information sampling, metacognition, and generalization in humans and macaques. *2021 – 2026*

Platt Lab, Wharton Neuroscience Initiative, University of Pennsylvania *Philadelphia, PA*
Eye-tracking, pupillometry, and drift-diffusion modeling of attention in consumer decision-making. *2020*

Junior Researcher Programme *Cambridge, UK*
Computational modeling of decision-making and intertemporal choice. *2020*

Judgment and Decision-Making Lab, University of Padua *Padua, Italy*
Behavioral experiments and modeling of judgment and decision-making for individual and societal well-being. *2018 – 2020*

Psicostat Lab, University of Padua *Padua, Italy*
Development of statistical methods for behavioral science. *2017 – 2020*

Awards & Honours

Excellence Award *UK*
For research contributions beyond expected grade level. Funded by the University of Oxford *2025*
£ 1,044

PhD Scholarship in Psychological Sciences *Italy*
Funded by the University of Padua *2020*
€ 45,343

Erasmus+ Scholarship, School of Psychology *Italy*
Funded by the University of Padua *2018*
€ 2,800

Talks & Conference Presentations

Mathematics Of Neuroscience and AI (Talk) *Rome, Italy*
Hybrid artificial neural network modelling predicts behaviour and neural activity in a solution for Buridan's ass. *2024*

Foraging and Information Seeking Conference (Poster) *Lyon, France*
Discovery of Cognitive Strategies for Information Sampling with Deep Cognitive Modelling and Investigation of their Neural Basis *2024*

Interdisciplinary Symposium on Decision Neuroscience (Poster) *Online*
Visuospatial working memory modulates the influence of visual attention on binary choice *2021*

Wharton Neuroscience Initiative Annual Meeting (Talk) *Online*
The effect of celebrity and visual attention on value-based decisions *2021*

Society for Neuroeconomics (Poster) *Online*
The Effect of Celebrity and Visual Attention on Value-Based Decisions *2020*

Italian Ministry of Economic Development – Division V (Talk) *Online*
Lessons from Decision Neuroscience to Inform Policy Making *2019*

Junior Researcher Programme Conference (Poster) *Siena, Italy*
The Attentional Drift Diffusion Model with Temporal Information *2019*

Psicostat Meeting (Talk) *Padua, Italy*
Estimation of the attentional drift diffusion model parameters via Random Utility *2019*

Teaching

Neuroanatomy, University of Oxford *Oxford, UK*
Undergraduate tutorial teaching for Prof. Matthew F. S. Rushworth. *2022*

Summer Schools

Computational Neuroscience, Neuromatch Academy *Online*
Project on fMRI decoding analysis supervised by Prof. Samuel J. Gershman *2021*

Psychology and Behavioral Sciences, JSchool *Siena, Italy*
Studying how to improve real-world decision making *2020*

Technical Skills & Methods

Computational Modeling	Neuroscience Methods	Programming & Infrastructure
Artificial neural networks – Symbolic regression – Reinforcement learning – Cognitive modelling – Bayesian statistics	Ultra-high field fMRI (7T) – Transcranial ultrasound stimulation (TUS) – Eye-tracking – Pupillometry – RSA – FSL – SPM	Python – Julia – R – JavaScript – HPC / SLURM – Git – LaTeX – Quarto

References

Prof. Matthew F. S. Rushworth
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Prof. Laurence T. Hunt
Department of Experimental Psychology
University of Oxford
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Prof. Michael L. Platt
Department of Neuroscience
University of Pennsylvania
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